A Simple LES Test of the Dynamic Wake Meandering Passive-Tracer Assumption

1 Basic Idea

The central assumption of the Dynamic Wake Meandering (DWM) model is that the transport of the wake deficit by large-scale turbulent motions can be described as a passive-tracer process. It implies that there exists a moving frame of reference, where the wake deficit is independent of the large-scale motions assumed to be responsible for wake displacement, although meandering has been associated with different mechanism[2]. Therefore, the aim is to test whether meandering can be treated as independent from wake evolution. Essentially, we will attempt to "unsmear" the effects of wake meandering using LES.

2 Background

For illustrative purposes, assume that the wake in the moving-frame of reference is Gaussian:

$$\Delta U_{mov}(y) = A \exp\left(-\frac{y^2}{2\sigma_{mov}^2}\right),$$

This is typically a reasonable approximation in the far wake, but please note, that DWM does **not** assume the wake to be Gaussian, as DWM originally proposed to use the Ainslie wake model[1].

If the wake-center displacement (meandering) is also Gaussian distributed,

$$\eta \sim \mathcal{N}\left(0, \sigma_{meander}^2\right).$$

which is often approximately observed for both the horizontal and vertical wake displacements when considered separately (show if this is the case). The time-averaged wake observed in the fixed frame is then:

$$\Delta U_{fix}(y) = \Delta U_{mov}(y) * p(\eta),$$

where $*$ denotes convolution.

The convolution of two Gaussians is again Gaussian,

$$\Delta U_{fix}(y) = A_{fix} \exp\left(-\frac{y^2}{2\sigma_{fix}^2}\right),$$

with

$$\sigma_{fix}^2 = \sigma_{mov}^2 + \sigma_{meander}^2.$$

Therefore, if we assume

1. the wake in the moving-frame is Gaussian,
2. the meandering distribution is Gaussian,
3. the wake behaves as a passive tracer,

then

$$\sigma_{mov} = \sqrt{\sigma_{fix}^2 - \sigma_{meander}^2}.$$

3 Gaussian Approximation Test

From LES, we can estimate:

- σ_{fix}: the standard deviation obtained from fitting a Gaussian profile to the fixed-frame wake deficit.
- $\sigma_{meander}$: the standard deviation of the wake-center position (as computed in a fixed coordinates). Note, there is no universal approach to calculating the wake center.

The DWM prediction for the width of the wake in the moving-frame of reference is therefore

$$\sigma_{mov,\mathrm{DWM}} = \sqrt{\sigma_{fix}^2 - \sigma_{meander}^2}.$$

This can be compared against the wake in the moving-frame of reference estimated from LES ($\sigma_{mov,\mathrm{LES}}$) by re-centering every wake realization, i.e. we can do a direct test of:

$$\sigma_{mov,\mathrm{LES}} \stackrel{?}{=} \sqrt{\sigma_{fix}^2 - \sigma_{meander}^2}.$$

Together, this can be reformulated into a quantifiable metric:

$$R = \frac{\sigma_{mov,\mathrm{LES}}}{\sqrt{\sigma_{fix}^2 - \sigma_{meander}^2}}.$$

The ratio directly isolates the Gaussian-convolution argument that underpins the passive-tracer interpretation of wake meandering. Hence, if $R \approx 1$, then the passive-tracer picture is consistent with LES. Significant deviations in R indicate inconsistency with either the Gaussian approximation, the passive-tracer assumption, or both. It should be noted that this test only compares the characteristic width of the wake. Therefore, $R \approx 1$ does not guarantee that the fixed-frame wake can be reproduced by a passive-tracer process. It merely indicates that the broadening associated with wake meandering is consistent with the Gaussian approximation.

4 Convolution Test: Direct Test of the Passive-Tracer Assumption

Rather than comparing only wake widths (or equivalently the second-order moments of the corresponding distributions), we can test the full convolution hypothesis. In contrast to the Gaussian approximation test (above), this approach does not assume a Gaussian wake deficit or a Gaussian meandering distribution. Instead, it relies directly on the passive-tracer assumption itself.

This test compares the fixed-frame wake obtained from LES against the convolution of the LES moving-frame wake with the measured probability distribution of wake displacements.

1. Estimate the wake in the moving-frame from LES (by re-centering the LES by the estimated meandering),
$$\Delta U_{mov,\mathrm{LES}}(y).$$

2. Estimate the probability density of wake position using LES,
$$p_{\mathrm{LES}}(\eta).$$

3. Predict the fixed-frame wake through convolution,
$$\Delta U_{fix,\mathrm{pred}} = \Delta U_{mov,\mathrm{LES}} * p_{\mathrm{LES}}.$$

4. Compare this prediction to the actual fixed-frame average wake obtained from LES.

Unlike the Gaussian approximation test, the convolution test does not require neither the wake deficit nor the meandering distribution to be Gaussian. Instead of comparing only a characteristic wake width, it compares the complete wake profile. The only requirement is the passive-tracer assumption itself, namely that individual wake realizations can be represented as translated copies of a unique moving-frame wake profile. Under the passive-tracer assumption, wake meandering acts only as a spatial translation of an otherwise unique wake profile. Consequently, the fixed-frame wake should be reproducible from two quantities only: (i) the moving-frame wake and (ii) the probability distribution of wake displacements.

It should be noted that the result depends on the definition of wake center used to construct the moving-frame reference system. The robustness of the conclusions could be evaluated with respect to different wake-center definitions, but I think that is out of scope. A simple approach would be to test different weigthing, i.e. $m = 1, 2, 3$.

If the DWM passive-tracer assumption is valid, the predicted and measured fixed-frame wake profiles should agree closely. Disagreement indicates that wake displacement and wake evolution cannot be treated independently. In such a situation, wake realizations cannot be represented as simple translations of a unique wake profile, implying a breakdown of the passive-tracer assumption.

References

[1] J. F. Ainslie. Calculating the flowfield in the wake of wind turbines. *Journal of Wind Engineering and Industrial Aerodynamics*, 27(1–3):213–224, 1988.

[2] X. Yang and F. Sotiropoulos. A review on the meandering of wind turbine wakes. *Energies*, 12(24), 2019.